



**COMSATS University Islamabad,
Abbottabad Campus**

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

< KOJLO >

Version 1.0

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

Supervised by
<Supervisor's Name>

Signature_____

Introduction

Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Khojlo** system. Khojlo is a mobile and web-based business discovery platform designed to help users find newly opened, affordable, and unique local businesses while providing business owners with a cost-effective way to increase their visibility. This document serves as a reference for developers, project supervisors, stakeholders, testers, and future maintainers throughout the development lifecycle.

Scope

Khojlo is a business discovery and comparison platform that enables users to explore, search, and compare local businesses through a structured and user-friendly interface. The system focuses on promoting newly established and value-driven businesses that often receive limited exposure on existing platforms.

The platform provides features such as user registration and profile management, business registration and management, discovery feeds for new and trending businesses, advanced search and filtering, business comparison, reviews and ratings, map-based location services, AI-powered chatbot assistance, and administrative moderation tools.

The primary goal of Khojlo is to bridge the gap between customers and business owners by improving business visibility, accessibility, and informed decision-making. The system is limited to business discovery and comparison and does not support online ordering, payment processing, or delivery services.

Overall description

Product perspective

Khojlo is a new mobile and web-based business discovery platform developed to address the limitations of existing business listing and recommendation systems. Current platforms such as Google Maps and Yelp primarily promote well-established and highly rated businesses, making it difficult for newly opened and lesser-known businesses to gain visibility.

The proposed system is not a replacement for any existing platform but rather an independent product that complements existing business discovery services by focusing on newly opened, affordable, unique, and value-driven businesses. Khojlo introduces features such as business comparison, structured discovery feeds, location-based search, and AI-assisted business information retrieval.

Operating environment

The Khojlo system will operate in the following environment:

OE-1

The system shall operate on Android smartphones running Android 10.0 or later.

OE-2

The system shall operate on modern web browsers including:

- *Google Chrome (latest versions)*
- *Mozilla Firefox (latest versions)*
- *Microsoft Edge (latest versions)*
- *Apple Safari (latest versions)*

OE-3

The frontend application shall be developed using Flutter and Dart and shall support both mobile and web platforms.

OE-4

The backend services shall be hosted on a cloud-based or dedicated server running a Linux operating system.

OE-5

The backend application shall be developed using Python 3.12 and FastAPI.

OE-6

The database shall be implemented using PostgreSQL 18.3 or later.

OE-7

The system shall require a stable internet connection for accessing business information, maps, reviews, and AI chatbot services.

OE-8

The system shall integrate with Google Maps SDK and Google Maps APIs for location tracking, navigation, and map visualization.

OE-9

Users, business owners, and administrators shall access the system remotely through the internet from their respective geographical locations.

Design and implementation constraints

CO-1

The frontend of the system shall be developed using Flutter 3.24+ and Dart 3.5+ as specified in the project proposal.

CO-2

The backend of the system shall be developed using Python 3.12 and FastAPI.

CO-3

The system shall use PostgreSQL as the primary database management system.

CO-4

The system shall communicate between frontend and backend components through RESTful APIs.

CO-5

The system shall use Google Maps APIs and Google Maps SDK for location-based services and navigation functionality.

CO-6

The AI chatbot module shall use a Retrieval-Augmented Generation (RAG) approach and shall only provide responses based on information stored within the platform database.

CO-7

The project shall follow Object-Oriented Design principles for system development and implementation.

CO-8

The project shall be developed using the Agile Software Development methodology with the Scrum framework.

CO-9

The system shall depend on internet connectivity and shall not support offline operation.

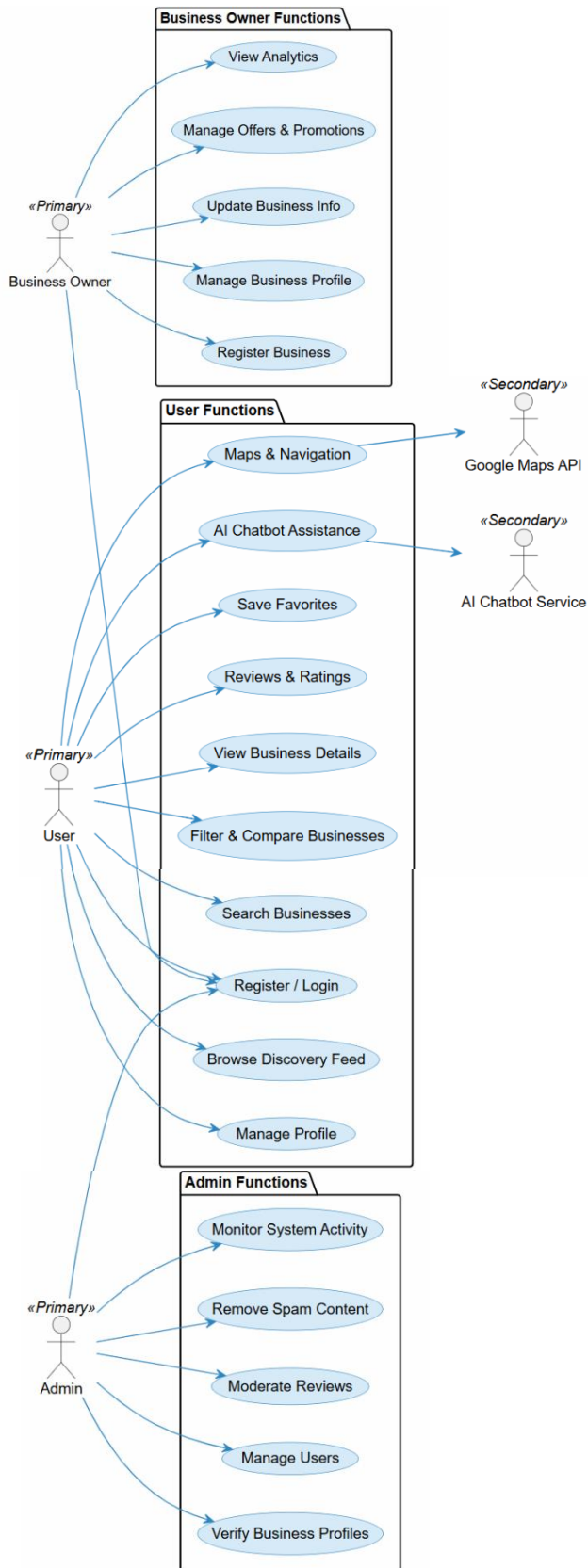
CO-10

The scope of the system shall be limited to business discovery, comparison, reviews, ratings, and information accessibility. Features such as online ordering, payment processing, and delivery services shall not be implemented.

Requirement identifying technique

The requirements for the Khojlo system were identified using the **Use Case Modeling** technique. This technique was selected because Khojlo is an interactive mobile and web-based application that involves direct interaction between different types of users and the system. Use cases help in understanding user goals, system behavior, and the functional requirements needed to achieve those goals.

Use case diagram



Use case description

Use cases are used to describe the interactions between system users and the Khojlo platform in order to achieve specific goals. They provide a structured method for capturing functional requirements by identifying the actors involved, the actions they perform, and the system responses generated during those interactions.

Table 1.1

Use Case ID:	UC-1
Use Case Name:	Register User Account
Actors:	Primary: User
Description:	A new user creates an account in the Khojlo system to access features like search, reviews, and saving businesses.
Trigger:	User clicks “Sign Up” option
Preconditions:	User is not already registered
Postconditions:	User account is created and stored in database
Normal Flow:	1. User selects sign up option 2. System displays registration form 3. User enters details 4. System validates data 5. System creates account 6. User is redirected to login/home page
Alternative Flows:	A1: User uses Google login; system creates account via OAuth
Exceptions:	E1: Email already exists, system shows error
Business Rules	BR-1: Email must be unique BR-2: Password must meet minimum security requirements
Assumptions:	User has valid email and internet access

Table 1.2

Use Case ID:	UC-2
Use Case Name:	Login User Account
Actors:	Primary: User
Description:	Registered user logs into the system.
Trigger:	User enters credentials
Preconditions:	User account exists
Postconditions:	User is authenticated and redirected to dashboard
Normal Flow:	1. User opens login page 2. User enters credentials 3. System validates credentials 4. System grants access

Alternative Flows:	A1: User logs in via Google OAuth
Exceptions:	E1: Invalid credentials
Business Rules:	BR-1: Credentials must match stored records
Assumptions:	User remembers credentials

Table 1.3

Use Case ID:	UC-3
Use Case Name:	Search and Filter Businesses
Actors:	Primary: User
Description:	User searches businesses using keywords and filters.
Trigger:	User enters search query
Preconditions:	Business data exists
Postconditions:	Matching businesses displayed
Normal Flow:	1. Enter keyword 2. System retrieves businesses 3. Apply filters 4. Display results
Alternative Flows:	A1: User applies multiple filters
Exceptions:	E1: No matching businesses found
Business Rules:	BR-1: Results ranked by relevance
Assumptions:	Internet available

Table 1.4

Use Case ID:	UC-4
Use Case Name:	View Business Details
Actors:	Primary: User
Description:	User views detailed information about a business.
Trigger:	User selects a business
Preconditions:	Business exists
Postconditions:	Business details displayed
Normal Flow:	1. User selects business 2. System retrieves profile 3. Display details 4. User interacts with options
Alternative Flows:	A1: User opens map view
Exceptions:	E1: Business data unavailable
Business Rules:	BR-1: Only verified business data displayed
Assumptions:	Business information exists

Table 1.5

Use Case ID:	UC-5
Use Case Name:	Submit Review and Rating
Actors:	Primary: User
Description:	User submits feedback for a business.
Trigger:	User clicks Add Review
Preconditions:	User logged in
Postconditions:	Review stored
Normal Flow:	1. Open business profile 2. Click Add Review 3. Enter rating/comment 4. Submit review 5. Store review
Alternative Flows:	A1: User edits review
Exceptions:	E1: Invalid review data
Business Rules:	BR-1: One review per user per business
Assumptions:	User has experience with business

Table 1.6

Use Case ID:	UC-6
Use Case Name:	Manage Business Profile
Actors:	Primary: Business Owner
Description:	Business owner manages business information.
Trigger:	Owner accesses dashboard
Preconditions:	Business registered
Postconditions:	Business profile updated
Normal Flow:	1. Login 2. Open profile 3. Edit information 4. Validate data 5. Save changes
Alternative Flows:	A1: Add promotions and offers
Exceptions:	E1: Invalid data entered
Business Rules:	BR-1: Only owner can edit profile
Assumptions:	Owner has authorization

Table 1.7

Use Case ID:	UC-7
Use Case Name:	View Business Discovery Feed
Actors:	Primary: User
Description:	User explores newly added and trending businesses.
Trigger:	User opens discovery feed
Preconditions:	Businesses available
Postconditions:	Feed displayed
Normal Flow:	1. Open feed 2. Retrieve businesses 3. Sort new/trending 4. Display feed
Alternative Flows:	A1: Refresh feed
Exceptions:	E1: No businesses available
Business Rules:	BR-1: New businesses prioritized
Assumptions:	Feed data exists

Table 1.8

Use Case ID:	UC-8
Use Case Name:	AI Chatbot Assistance
Actors:	Primary: User
Description:	User asks chatbot business-related questions.
Trigger:	User opens chatbot
Preconditions:	Business data available
Postconditions:	Response provided
Normal Flow:	1. Open chatbot 2. Enter question 3. Process query 4. Retrieve data 5. Respond
Alternative Flows:	A1: Unsupported question
Exceptions:	E1: No matching information found
Business Rules:	BR-1: Responses based on platform data
Assumptions:	Internet available

Table 1.9

Use Case ID:	UC-9
Use Case Name:	View Business on Map
Actors:	Primary: User
Description:	User views business location through maps.
Trigger:	User clicks View on Map
Preconditions:	Business has location data
Postconditions:	Map displayed
Normal Flow:	1. Select map option 2. Load map 3. Show location 4. Allow navigation
Alternative Flows:	A1: Request directions
Exceptions:	E1: Location unavailable
Business Rules:	BR-1: Valid GPS coordinates required
Assumptions:	Google Maps service available

Table 1.10

Use Case ID:	UC-10
Use Case Name:	Admin Moderation
Actors:	Primary: Admin
Description:	Admin manages and moderates platform content.
Trigger:	Admin logs into dashboard
Preconditions:	Admin privileges granted
Postconditions:	Platform updated
Normal Flow:	1. Login 2. View dashboard 3. Verify businesses 4. Moderate content 5. Update records
Alternative Flows:	A1: Block user account
Exceptions:	E1: Unauthorized access
Business Rules:	BR-1: Only admin can moderate content
Assumptions:	Admin credentials valid

Functional Requirements

The functional requirements describe the services and functions that the Khojlo system shall provide to its users. These requirements are organized according to the major features of the system.

Feature 1: User Authentication and Profile Management

This feature enables users to register, log in, and manage their profiles.

Identifier	FR-1
Title	User Registration
Requirement	The user shall be able to create a new account using email and password.
Source	User
Rationale	Allow new users to access the platform.
Business Rule	BR-1 Email must be unique
Dependencies	None
Priority	High

Identifier	FR-2
Title	User Login
Requirement	The user shall be able to log into the system using valid credentials.
Source	User
Rationale	Provide secure access to platform services.
Business Rule	BR-2 Credentials must be validated
Dependencies	FR-1

Priority	High
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Feature 2: Business Discovery and Search

Users can discover, search, filter and view detailed information about businesses.

Identifier	FR-3
Title	Search Businesses
Requirement	The user shall be able to search businesses using keywords and categories.
Source	User
Rationale	Enable business discovery.
Business Rule	None
Dependencies	FR-2
Priority	High

Identifier	FR-4
Title	Filter Businesses
Requirement	The user shall be able to filter businesses by category, location and price.
Source	User
Rationale	Improve search accuracy.
Business Rule	None
Dependencies	FR-3

Priority	High
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Identifier	FR-5
Title	View Business Details
Requirement	The system shall display complete business information.
Source	User
Rationale	Provide informed decision making.
Business Rule	BR-3 Only verified business profiles displayed
Dependencies	FR-3
Priority	High

Feature 3: User Reviews and Favorites

Users can leave reviews, submit ratings, and save favorite businesses for quick access.

Identifier	FR-6
Title	Submit Reviews
Requirement	The user shall be able to submit ratings and reviews.
Source	User
Rationale	Support transparency and trust.
Business Rule	BR-4 One review per business per user
Dependencies	FR-2
Priority	High

Identifier	FR-7
Title	Save Favorites
Requirement	The user shall be able to save businesses to favorites.
Source	User
Rationale	Allow quick future access.
Business Rule	None
Dependencies	FR-2
Priority	Medium

Feature 4: Business Owner Management

Business owners can register their businesses, manage profiles, and create promotional offers.

Identifier	FR-8
Title	Business Registration
Requirement	Business owners shall be able to register businesses.
Source	Business Owner
Rationale	Allow businesses to join platform.
Business Rule	BR-5 Business verification required
Dependencies	FR-2
Priority	High

Identifier	FR-9
Title	Manage Business Profile
Requirement	Business owners shall be update business information.
Source	Business Owner
Rationale	Maintain accurate information.
Business Rule	BR-5 Business verification required
Dependencies	FR-8
Priority	High

Identifier	FR-10
Title	Manage Promotions
Requirement	Business owners shall create promotional offers.
Source	Business Owner
Rationale	Increase business visibility.
Business Rule	None
Dependencies	FR-9
Priority	Medium

Feature 5: System Features

Core system features including discovery feed, map integration, and AI chatbot support.

Identifier	FR-11
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Title	Discovery Feed
Requirement	The system shall display new and trending businesses.
Source	System
Rationale	Promote new businesses.
Business Rule	BR-6 New businesses prioritized
Dependencies	FR-8
Priority	High

Identifier	FR-12
Title	Map Integration
Requirement	The system shall display business locations on maps.
Source	User
Rationale	Assist navigation.
Business Rule	BR-7 Valid GPS coordinates required
Dependencies	FR-5
Priority	High

Identifier	FR-13
Title	AI Chatbot
Requirement	The system shall answer business-related queries using RAG.

Source	User
Rationale	Provide instant assistance.
Business Rule	BR-8 Responses based on stored platform data
Dependencies	FR-5
Priority	High

Feature 6: Admin and Moderation

Administrative functions for verifying business profiles and moderating platform content.

Identifier	FR-14
Title	Admin Verification
Requirement	Admin shall verify business profiles before publication.
Source	Admin
Rationale	Ensure authenticity.
Business Rule	BR-9 Only admins may verify businesses
Dependencies	FR-8
Priority	High

Identifier	FR-15
Title	Content Moderation
Requirement	Admin shall remove spam and inappropriate content.
Source	Admin

Rationale	Maintain platform quality.
Business Rule	BR-10 Moderation policies enforced
Dependencies	FR-14
Priority	High

Non-Functional Requirements

Usability

The usability requirements ensure that users can easily learn and interact with the Khojlo system.

USE-1: *The system shall allow users to search for businesses using a maximum of three interactions from the home screen.*

USE-2: *The system shall provide a consistent user interface across mobile and web platforms.*

USE-3: *The system shall display clear validation messages whenever users enter invalid data.*

USE-4: *The system shall allow users to access business details directly from search results with a single click/tap.*

USE-5: *The AI chatbot interface shall always be accessible from the main navigation menu.*

USE-6: *New users shall be able to complete account registration within 2 minutes under normal conditions.*

Performance

The performance requirements define the expected speed and responsiveness of the system.

PER-1: *95% of application pages shall load completely within 4 seconds over a 20 Mbps or faster internet connection.*

PER-2: *The system shall return search results within 3 seconds for 95% of search requests.*

PER-3: *The system shall display business profile details within 2 seconds after user selection.*

PER-4: *The AI chatbot shall respond to user queries within 5 seconds for 90% of requests.*

PER-5: *The system shall support at least 200 concurrent users without significant degradation in performance.*

Reliability

The system shall provide dependable and consistent operation.

REL-1: *The system shall maintain data integrity during all create, update, and delete operations.*

REL-2: *The system shall automatically recover from temporary server failures without loss of stored data.*

REL-3: *The system shall successfully process at least 99% of valid user requests.*

REL-4: *User reviews, ratings, and business information shall remain persistent after system restarts.*

Security

The system shall protect user data and prevent unauthorized access.

SEC-1: *All user passwords shall be stored in encrypted form.*

SEC-2: *Only verified business owners shall be allowed to modify business profiles.*

SEC-3: *Administrative functions shall be accessible only to authorized administrators.*

SEC-4: *The system shall use secure HTTPS communication for all client-server interactions.*

Scalability

The system shall be capable of handling future growth.

SCA-1: *The system architecture shall support the addition of new business categories without requiring major modifications.*

SCA-2: *The system shall support future expansion to additional cities and geographical regions.*

SCA-3: *The database shall support growth in the number of users, businesses, reviews, and ratings without significant performance degradation.*

References

List any documents or other resources to which this SRS refers, if any. These might include user interface style guides, standards, system requirements specifications, interface specifications, or the SRS for a related product.